

# NUMBER PATTERNS

Short Revision Notes — Evaluate Level (Make Judgments and Justify Decisions)

## 1. Verifying Whether a Value Is a Term

| Question                         | Method                            | Verification                    |
|----------------------------------|-----------------------------------|---------------------------------|
| Is 75 a term of $T_n = 4n - 3$ ? | Set $4n - 3 = 75$                 | $4n = 78 \rightarrow n = 19.5$  |
| Judgment                         | Is $n$ a whole number?            | NO (19.5 is not a whole number) |
| Conclusion                       | 75 is NOT a term of this sequence | ✓                               |

| Question                         | Method                      | Verification                 |
|----------------------------------|-----------------------------|------------------------------|
| Is 0 a term of $T_n = 56 - 4n$ ? | Set $56 - 4n = 0$           | $4n = 56 \rightarrow n = 14$ |
| Judgment                         | Is $n$ a whole number?      | YES (14 is a whole number)   |
| Conclusion                       | 0 IS a term (the 14th term) | ✓                            |

## 2. Checking the Correctness of a General Term

| Given Sequence    | Proposed $T_n$  | Verification                                                    | Judgment |
|-------------------|-----------------|-----------------------------------------------------------------|----------|
| 5, 8, 11, 14...   | $T_n = 5n + 3$  | $T_1 = 5(1)+3 = 8 \times$                                       | WRONG    |
| 5, 8, 11, 14...   | $T_n = 3n + 2$  | $T_1 = 3(1)+2 = 5 \checkmark$ , $T_2 = 3(2)+2 = 8 \checkmark$   | CORRECT  |
| 10, 17, 24, 31... | $T_n = 7n + 3$  | $T_1 = 7(1)+3 = 10 \checkmark$ , $T_2 = 7(2)+3 = 17 \checkmark$ | CORRECT  |
| 22, 19, 16, 13... | $T_n = 25 - 3n$ | $T_1 = 25-3 = 22 \checkmark$ , $T_2 = 25-6 = 19 \checkmark$     | CORRECT  |
| 22, 19, 16, 13... | $T_n = 22 - 3n$ | $T_1 = 22-3 = 19 \times$                                        | WRONG    |

**Justification Method:** Find the common difference ( $d$ ) · The coefficient of  $n$  MUST equal  $d$  · Substitute  $n = 1$  to check  $T_1$  matches · Substitute  $n = 2$  to check  $T_2$  matches

## 3. Judging Whether a Sequence Is Correct

| Statement                                                  | Check                             | Judgment                   |
|------------------------------------------------------------|-----------------------------------|----------------------------|
| "The sequence 2, 4, 6, 8... has a common difference"       | $4-2 = 2$ , $6-4 = 2$ , $8-6 = 2$ | TRUE ( $d = 2$ ) ✓         |
| "The sequence 1, 1, 2, 2, 3, 3... has a common difference" | $1-1 = 0$ , $2-1 = 1$ , $2-2 = 0$ | FALSE (differences vary) ✗ |
| "The sequence 3, 3, 3, 3... has a common difference"       | $3-3 = 0$ , $3-3 = 0$             | TRUE ( $d = 0$ ) ✓         |

**Justification:** For a common difference sequence, the difference between consecutive terms must ALWAYS be the same.

## 4. Comparing Two Sequences — Which Is Better?

| Sequence A          | Sequence B           | Comparison |
|---------------------|----------------------|------------|
| 5, 8, 11, 14, 17... | 5, 11, 17, 23, 29... |            |
| $T_n = 3n + 2$      | $T_n = 6n - 1$       |            |
| $d = 3$             | $d = 6$              |            |

**Evaluation:**

| Criteria          | Sequence A    | Sequence B     |
|-------------------|---------------|----------------|
| Common difference | 3             | 6              |
| Growth rate       | Slower        | Faster         |
| 20th term         | $T_{20} = 62$ | $T_{20} = 119$ |
| Which is larger?  | Smaller       | LARGER         |

**Judgment:** If you need LARGER terms quickly → choose Sequence B. If you need slower growth → choose Sequence A.

## 5. Judging Student Mistakes

| Student's Work                                          | Analysis                            | Judgment                                                   |
|---------------------------------------------------------|-------------------------------------|------------------------------------------------------------|
| "For 5, 8, 11, 14, the general term is $T_n = 5n + 3$ " | $T_1 = 5(1)+3 = 8$ (should be 5)    | INCORRECT – The student used the first term as coefficient |
| "For 22, 19, 16, 13, the common difference is +3"       | $19-22 = -3$ (not +3)               | INCORRECT – Student forgot to subtract in correct order    |
| "For 3, 6, 9, 12, $T_n = 3n$ "                          | $T_1 = 3(1) = 3$ , $T_2 = 3(2) = 6$ | CORRECT – Student identified $d=3$ and $c=0$               |

### Common Mistakes to Judge:

| Mistake                            | Why It's Wrong                                   | Correct Version                    |
|------------------------------------|--------------------------------------------------|------------------------------------|
| $T_n = \text{first term} \times n$ | Coefficient is common difference, not first term | $T_n = \text{first term} + (n-1)d$ |
| $T_n = T_1 + (n+1)d$               | Formula uses $(n-1)$ , not $(n+1)$               | $T_n = T_1 + (n-1)d$               |
| For 10, 7, 4, 1: $T_n = 10 - 3$    | Missing the $n$ term                             | $T_n = 13 - 3n$                    |

## 6. Evaluating "Term Existence" Problems

| Problem                            | Test                                             | Verdict        |
|------------------------------------|--------------------------------------------------|----------------|
| Is 15 a term of $T_n = 3n + 6$ ?   | $3n+6=15 \rightarrow 3n=9 \rightarrow n=3$       | YES (3rd term) |
| Is 50 a term of $T_n = 4n - 3$ ?   | $4n-3=50 \rightarrow 4n=53 \rightarrow n=13.25$  | NO (not whole) |
| Is -5 a term of $T_n = 20 - 5n$ ?  | $20-5n=-5 \rightarrow 5n=25 \rightarrow n=5$     | YES (5th term) |
| Is -20 a term of $T_n = 30 - 4n$ ? | $30-4n=-20 \rightarrow 4n=50 \rightarrow n=12.5$ | NO (not whole) |

**Justification Pattern:** If  $n$  is a positive integer  $\rightarrow$  YES, it is a term. If  $n$  is not a positive integer  $\rightarrow$  NO, it is not a term. If  $n = 0$  or negative  $\rightarrow$  NO, terms start from  $n=1$ .

## 7. Evaluating Real-World Reasonableness

| Situation                        | $T_n$ Formula                              | Evaluation                                         |
|----------------------------------|--------------------------------------------|----------------------------------------------------|
| A runner adds 100m daily         | $T_n = 500 + (n-1)100$                     | REASONABLE ✓                                       |
| "Day 100 runner runs 10,400m"    | $T_{100} = 500 + 99(100) = 10,400\text{m}$ | UNREASONABLE (10.4km daily! Too high for training) |
| "Table has 50 seats in 15th row" | $T_{15} = 9 + 14(3) = 51$ seats            | UNREASONABLE (rounding error – should be 51)       |

**Judgment Criteria:** Does the formula match the given pattern? ✓ Does the computed value make sense in context? ✓  
Are the units correct? ✓

## 8. Judging Which Sequence Is Correct From Data

| Data                         | Proposed Sequence A | Proposed Sequence B | Evaluation |
|------------------------------|---------------------|---------------------|------------|
| $T_1 = 5, T_2 = 9, T_3 = 13$ | $T_n = 4n + 1$      | $T_n = 5n - 1$      |            |
| Check $T_1$                  | $4(1)+1 = 5$ ✓      | $5(1)-1 = 4$ ✗      |            |
| Check $T_2$                  | $4(2)+1 = 9$ ✓      | $5(2)-1 = 9$ ✓      |            |
| Check $T_3$                  | $4(3)+1 = 13$ ✓     | $5(3)-1 = 14$ ✗     |            |
| Judgment                     | CORRECT ✓           | INCORRECT ✗         |            |

## 9. Evaluating Term Behavior

| $T_n$           | When do terms become positive?                        | When do terms become negative? | Judgment                                       |
|-----------------|-------------------------------------------------------|--------------------------------|------------------------------------------------|
| $T_n = 3n - 10$ | $3n-10 > 0 \rightarrow n > 3.33 \rightarrow n \geq 4$ | $n < 4$                        | Terms 1-3 are negative; from 4th term positive |
| $T_n = 50 - 7n$ | $50-7n < 0 \rightarrow n > 7.14 \rightarrow n \geq 8$ | $n > 7.14$                     | 8th term and beyond are negative               |
| $T_n = 4n + 5$  | Always positive                                       | Never                          | All terms are positive                         |

**Evaluation Checklist:** At what term does sign change occur? · Is the threshold a whole number? · How many positive/negative terms exist?

## 10. Evaluating Method Efficiency

| Task                   | Method A                 | Method B            | Which is Better?             |
|------------------------|--------------------------|---------------------|------------------------------|
| Find 100th term        | Write 100 terms          | Use $T_n$ formula   | Method B – saves time ✓      |
| Check if 15 is a term  | Check each term until 15 | Use $T_n$ formula   | Method B – faster ✓          |
| Find common difference | $T_3 - T_2$              | $T_2 - T_1$         | Both – either gives same d   |
| Find $T_1$             | Substitute n=1           | Count from sequence | Substitution – more accurate |

**Evaluation:** For long sequences → Formula method is SUPERIOR. For short sequences → Manual counting may be acceptable. For accuracy → Always use formula to verify.

## 11. Evaluating Real-World Word Problems

| Problem                               | $T_n$ Form              | Evaluation                     |
|---------------------------------------|-------------------------|--------------------------------|
| Seats: Row 1=9, Row 2=12, Row 3=15... | $T_n = 3n + 6$          |                                |
| "Row 10 has 36 seats"                 | $T_{10} = 3(10)+6 = 36$ | CORRECT ✓                      |
| "Row 15 has 51 seats"                 | $T_{15} = 3(15)+6 = 51$ | CORRECT ✓                      |
| "There are 120 seats in row 30"       | $T_{30} = 3(30)+6 = 96$ | INCORRECT (overestimated)<br>✗ |

| Problem                                      | $T_n$ Form                 | Evaluation                     |
|----------------------------------------------|----------------------------|--------------------------------|
| Money: Week 1=100, Week 2=110, Week 3=120... | $T_n = 10n + 90$           |                                |
| "Week 10 has Rs 180"                         | $T_{10} = 10(10)+90 = 190$ | INCORRECT (should be Rs 190) ✗ |
| "Week 20 has Rs 290"                         | $T_{20} = 10(20)+90 = 290$ | CORRECT ✓                      |

## 12. Judging Reasonableness of Findings

| Claim                                       | Check                       | Is it Reasonable?                           |
|---------------------------------------------|-----------------------------|---------------------------------------------|
| "The 100th term of 5, 8, 11, 14... is 302"  | $T_{100} = 3(100)+2 = 302$  | YES, it's reasonable ✓                      |
| "The 50th term of 1, 7, 13, 19... is 300"   | $T_{50} = 6(50)-5 = 295$    | NO, it should be 295 (overestimated by 5) ✗ |
| "The 15th term of 100, 95, 90, 85... is 30" | $T_{15} = 105 - 5(15) = 30$ | YES, it's reasonable ✓                      |

**Reasonableness Checks:** Is the sequence increasing or decreasing? · Does the term value follow the pattern? · Is the magnitude reasonable for the term position?

## 13. Comparing Different Sequence Patterns

| Pattern A                       | Pattern B               | Evaluation   |
|---------------------------------|-------------------------|--------------|
| 2, 4, 6, 8, 10...               | 2, 4, 8, 16, 32...      |              |
| $T_n = 2n$                      | $T_n = 2^n$             |              |
| $T_5 = 10$                      | $T_5 = 32$              | (Comparison) |
| $T_{10} = 20$                   | $T_{10} = 1024$         | (Comparison) |
| Arithmetic pattern grows slowly | Exponential grows fast  | (Judgment)   |
| Better for simple counting      | Better for rapid growth | (Judgment)   |

## 14. Evaluation Checklist

**When verifying a general term:**

- Does  $T_1$  match the first term?
- Does  $T_2$  match the second term?
- Does  $T_3$  match the third term?
- Is the coefficient of  $n$  equal to the common difference?

**When checking if a value is a term:**

- Set  $T_n = \text{value}$
- Solve for  $n$
- Is  $n$  a positive integer? Yes → It IS a term. No → It IS NOT a term.

**When comparing sequences:**

- What is the common difference?
- Which grows faster?
- Which is more practical?

**When evaluating word problems:**

- Does the answer make sense?
- Are the units correct?
- Is the value reasonable?

**When checking student work:**

- Did they find  $d$  correctly?
- Did they use the correct formula?
- Did they verify with  $n=1$  and  $n=2$ ?
- Did they check if  $n$  is a whole number?

## 15. Self-Test — Evaluation Questions

| No. | Question                                                 | Answer & Justification                                                            |
|-----|----------------------------------------------------------|-----------------------------------------------------------------------------------|
| 1   | Is $T_n = 4n + 5$ correct for 9, 13, 17, 21?             | $T_1 = 4(1)+5=9$ ✓, $d=4$ ✓ → CORRECT                                             |
| 2   | Is 85 a term of $T_n = 6n + 1$ ?                         | $6n+1=85 \rightarrow n=14 \rightarrow$ YES (14th term)                            |
| 3   | Is 100 a term of $T_n = 7n - 5$ ?                        | $7n-5=100 \rightarrow n=15 \rightarrow$ YES (15th term)                           |
| 4   | Which is better: $T_n=3n+1$ or $T_n=3n+4$ for 4,7,10,13? | $T_n=3n+1 \rightarrow T_1=4$ ✓; $T_n=3n+4 \rightarrow T_1=7$ ✗ → FIRST is correct |
| 5   | "A runner covers 10,000m on day 50" – reasonable?        | $T_{50}=500+49(100)=5,400\text{m} \rightarrow 10,000\text{m}$ is UNREASONABLE ✗   |
| 6   | Student says $d=5$ for 20, 15, 10, 5. Is this correct?   | $15-20 = -5 \rightarrow$ INCORRECT ( $d=-5$ ) ✗                                   |
| 7   | Is -10 a term of $T_n = 30 - 4n$ ?                       | $30-4n=-10 \rightarrow n=10 \rightarrow$ YES (10th term)                          |
| 8   | Which grows faster: $T_n=5n$ or $T_n=3n$ ?               | $T_n=5n$ ( $d=5$ ) grows FASTER than $T_n=3n$ ( $d=3$ )                           |
| 9   | Student says $T_n=20n$ for 20, 40, 60. Correct?          | $T_1=20(1)=20$ ✓, $T_2=20(2)=40$ ✓ → CORRECT                                      |
| 10  | Is 50 a term of $T_n=8n-3$ ?                             | $8n-3=50 \rightarrow n=6.625 \rightarrow$ NO (not whole number)                   |

## 16. Evaluation Decision Matrix

| Situation                     | Decision                         |
|-------------------------------|----------------------------------|
| $T_n$ matches $T_1, T_2, T_3$ | ACCEPT the general term          |
| $T_n$ doesn't match $T_1$     | REJECT – adjust the constant     |
| $T_n$ doesn't match $T_2$     | REJECT – check common difference |
| $n$ is a whole number         | Value IS a term                  |
| $n$ is not a whole number     | Value IS NOT a term              |
| $d > 0$                       | Sequence is INCREASING           |
| $d < 0$                       | Sequence is DECREASING           |
| $d = 0$                       | Sequence is CONSTANT             |

**Revision Tip:** When evaluating, always ask "Does this make sense?" Justification is about proving WHY something is correct or incorrect – not just saying "yes" or "no"!